

SAFAR PAHAD PARIVAR · VIDEO KIT

Kit User Guide

Setup, titles, GPS, captions, route maps, sound effects, music, moments and rendering — in the order you make a video.

सफ़र पहाड़ परिवार — वीडियो बनाने की पूरी गाइड

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Everything in the kit, in the order you use it when making a video. Menu scripts live in **Workspace** → **Scripts** → **Safar Pahad Parivar**; titles live in **Effects** → **Titles** → **Safar Pahad Parivar**.

```
Workspace → Scripts → Safar Pahad Parivar
├─ 1 Setup
│   │   Setup Render Presets · New Timeline - YouTube 16x9
│   │   New Timeline - Shorts 9x16 · Import Brand Graphics
│   │   Keyboard Shortcuts - On or Off
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│   │   Moments - Add Markers / Best Moments Timeline / Search Transcript
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├─ 6 Sound & Music
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│   │   SFX - Auto Sound for Titles / Remove Auto Sounds
│   │   SFX - Land at Playhead / Loop Fill (In to Out)
│   │   Music - Key Transition · Distance - Selected Clips
```

0. Where each tool runs — inside Resolve or outside

Almost everything is used inside DaVinci Resolve. Outside Resolve you only run a few PowerShell commands — once per PC to install/build things, and at the start of each trip to make folders and sort footage — and you read two text reports the moment finder writes.

Every menu script in detail — what you need first, what changes in Resolve after running it, how to redo / undo: [SPP_Menu_Scripts.md](#) . **New project?** Read its section A first — don't import the kit folder into the Media Pool.

Inside Resolve — menu scripts (**Workspace** → **Scripts** → **Safar Pahad Parivar** → **folder** → **script**)

The scripts sit in six folders, in the order you use them while making a video:

Menu folder	Menu script	What you select first	What it does
1 Setup	Setup Render Presets (once per PC)	—	adds the SPP YouTube 4K / 1080p / Shorts render presets
	New Timeline - YouTube 16x9 / Shorts 9x16	—	named tracks (Main, B-Roll, GFX, Nat Sound, VO, Music, SFX, Subtitles)
	Import Brand Graphics	—	intro, end card, watermarks into a bin
	Keyboard Shortcuts - On or Off	—	number keys 4–0 tag clips; Ctrl+Alt+B / M / S import brand graphics / music / SFX
2 Marking & Moments	Tag Shot Type → 0–9	timeline or media-pool clips	colour + keyword (Hero, Family talk, Scenery, Road, Drone, Timelapse, Kids, Stay/Food, Reject)

Menu folder	Menu script	What you select first	What it does
	Moments - Analyse Trip	trip footage in the project	finds laughter / kids / reactions + Hindi transcript (background)
	Moments - Add Markers · Best Moments Timeline · Search Transcript	—	markers on clips · a selects timeline · a timeline of every clip where a word was said
3 Picture	Reflection - Scan Trip	trip footage in the project	colours the clips with windshield / window reflections
	Reflection - Clean Selected Clips	video clips shot through the glass	removes dashboard / phone reflections → cleaned take on the same clip (background)
	Reflection - Show Original or Cleaned	cleaned clips (or none = all)	before / after switch
4 Titles & Graphics	Info Cards - Fill from GPS	— (all empty SPP Info Cards)	place, altitude, date, time, weather from the GPS of the shot underneath
	Route Map - Build from Timeline	an SPP Route Map clip on the timeline	real-road route, stops, times for the trip
	Chapters - Number + YouTube List	SPP Chapter titles on the timeline	numbers the chapters, adds markers, copies the YouTube chapter list
	Titles - Follow Clip Length	— (all SPP titles)	entrance fixed, exit always on the clip's last frame, hold stretches (once; then automatic)
5 Captions & Voice	Captions - Sync Words to VO	subtitles + a track named VO	word-by-word timing + stress into SPP Captions
	Phone Voice - Selected Clips	audio clips	phone / walkie-talkie voice
6 Sound & Music	SFX - Import Library · Music - Import Library	—	puts the libraries into the <i>SPP SFX</i> / <i>SPP Music</i> bins
	SFX - Auto Sound for Titles · SFX - Remove Auto Sounds	—	beat-timed sounds for every SPP title (Strings / Grand / Light / Mix) · clears them
	SFX - Land at Playhead	one sound in the Media Pool	its hit / swell lands exactly on the playhead
	SFX - Loop Fill (In to Out)	one <code>_LOOP_</code> sound + In/Out	repeats it seamlessly for any length
	Music - Key Transition	1 or 2 music clips on the timeline	detects the keys, places a bridge / swell / tail on the cut
	Distance - Selected Clips	audio clips	near / mid / far / very far / across the valley

Titles are in **Effects** → **Titles** → **Safar Pahad Parivar**: Film Title, Chapter, Info Card, Altitude Counter, Peak Callout, Pop-up Title, Route Map, Captions, Credits (§6–§13). Colour tool **SPP Glass Glare** (takes the milky glare off windshield shots) is on the Color page: **ResolveFX Color** → **DCTL** → *SPP Glass Glare* (§15e). Script output and progress show in **Workspace** → **Console**.

Outside Resolve — PowerShell in the kit folder (F:\Video Editing_Safar Pahad Parivar Kit)

When	Command	Time
New PC / after git pull of kit changes	<code>.\install.ps1</code> then restart Resolve	seconds
New PC (once)	<code>.\Tools\setup_word_timing.ps1</code> — installs all Python engines (speech, sound events, GPU)	10–20 min
New PC (once) or when the guide says the library changed	<code>.\Tools\make_sfx.ps1</code> — builds SFX\ (downloads the instruments + real recordings the first time)	15–25 min first time, ~10 min later
New PC (once) or when the music changed	<code>.\Tools\make_music.ps1</code> (one track: <code>.\Tools\make_music.ps1 Pahadi</code>)	~10 min
New trip / video	<code>.\Tools\new_video.ps1 -Trip "2026-10 Chopta Tungnath" -Video "01 Main Film"</code>	seconds
New trip	<code>python Tools\spp_sort_media.py "<trip>\Footage" --apply</code> (sort by orientation)	a minute
Only to fetch more real recordings	<code>Tools\.venv\Scripts\python.exe Tools\spp_freesound.py</code> (needs <code>Tools\freesound_key.txt</code>)	minutes

Outside Resolve — files you read

File	What
<code><trip>_spp_moments\Moments.md</code>	3 opening candidates, top 40 moments, day-by-day list (after <i>Moments - Analyse Trip</i>)
<code><trip>_spp_moments\Transcript.md</code>	everything said on the trip, clip by clip (Ctrl+F)
<code><trip>\stops.csv</code>	route-map stops — edit names/times, then run <i>Route Map - Build from Timeline</i> again
<code>Docs\SFX_Library.md</code> · <code>Docs\Music_Library.md</code>	every sound / track, what it's for, length, key
<code><trip>_spp_clean\reflection_sheet.jpg</code> · <code>reflection_report.csv</code>	clips with reflections (after <i>Reflection - Scan Trip</i>)
<code><trip>_spp_clean<clip>__f...jpg</code>	before / after / removed picture for each cleaned clip
<code>Docs*.mp3</code>	demo reels: listen before choosing

Optional command-line tools (the menu scripts call these for you)

`spp_gps.py` (GPS lookup, stops, route, GPX, local API) · `spp_key.py` (find a song's key / make a transition) · `spp_distance.py` · `spp_phone_voice.py` · `spp_moments.py --search "बर्फ"` · `spp_reflection.py preview|clean|scan "<clip or trip>"` — all run with `Tools\.venv\Scripts\python.exe` .

A video, start to finish

1. **PowerShell:** `new_video.ps1` → copy footage into `<trip>\Footage` → `spp_sort_media.py --apply` .

2. **Resolve:** new project → import footage → **New Timeline** → **Import Brand Graphics** → **Moments - Analyse Trip** (and **Reflection - Scan Trip** if you filmed through the car windows).
3. Read `Moments.md` ; **Moments - Add Markers / Best Moments Timeline; Tag Shot Type** while you watch.
4. Edit. Intro → **SPP Film Title** right after it; an **SPP Chapter** card at the start of each chapter. Shots through the glass: **Reflection - Clean Selected Clips** (+ *SPP Glass Glare* when grading). Add SPP titles → **Info Cards - Fill from GPS, Route Map - Build from Timeline**.
5. Record the VO onto the **VO** track → subtitles → **Captions - Sync Words to VO**.
6. Music from *SPP Music* → **Music - Key Transition** where tracks change. **SFX - Auto Sound for Titles**, real sounds from folders 20–24 (use **Loop Fill** for beds, **Distance** for far-away sounds, **Land at Playhead** for risers).
7. Mix (VO recipe, §15) → render with an SPP preset → save to `<video>\Exports` .
8. **Chapters - Number + YouTube List** → paste the chapter list into the YouTube description.

1. One-time setup

Do this on a new or rebuilt PC (or after pulling kit changes).

Step	What to do
1	Install DaVinci Resolve Studio, Git + Git LFS , Python 3.13 from python.org (not the Microsoft Store one), <code>ffmpeg</code> (<code>choco install ffmpeg</code>). An NVIDIA GPU is used for speech (captions, moments).
2	<code>git clone https://github.com/pushpenderpannu/Safar-Pahad-Parivar-Kit.git "F:\Video Editing_Safar Pahad Parivar Kit"</code> then <code>git lfs pull</code>
3	PowerShell in the kit folder: <code>.\install.ps1</code> — copies menu scripts + titles into Resolve.
4	<code>.\Tools\setup_word_timing.ps1</code> — installs all Python engines: caption timing, GPS, route maps, sound, music, key finder, moment finder (incl. its 312 MB sound-event model; the 3 GB speech model downloads on first use).
5	Optional, for the real-world sounds: create a free key at freesound.org/apiv2/apply and save it as <code>Tools\freesound_key.txt</code> (git ignores it). Without it the SFX library is built without folders 20–24.
6	<code>.\Tools\make_sfx.ps1</code> — builds the sound-effects library (first time 15–25 min: downloads ~1.1 GB of CC0 instrument recordings and the real-world recordings).
7	<code>.\Tools\make_music.ps1</code> — builds the 12 background-music tracks (~10 min).
8	Restart Resolve, then once: 1 Setup → Setup Render Presets, 6 Sound & Music → SFX - Import Library and Music - Import Library (menu scripts).

Disk space: ~5 GB for the generated libraries and the instrument recordings (`SFX` 2 GB, `Music` 0.5 GB, `Source\_vsco` 2 GB, `Source\_cc0` 0.3 GB) — none of it is in git.

After any kit update: `git pull` → `.\install.ps1` → restart Resolve. Re-run `make_sfx.ps1` / `make_music.ps1` only when the update changed the sounds or music (the commit message says so).

2. Start a new trip / video

One folder per **trip** (footage stored once), one sub-folder + one Resolve project per **video**.

```
.\Tools\new_video.ps1 -Trip "2026-10 Chopta Tungnath" -Video "01 Main Film"
```

Copy the camera/phone files into `<trip>\Footage`, then sort them by real orientation:

```
python Tools\spp_sort_media.py "<trip>\Footage" # dry run - shows the plan
python Tools\spp_sort_media.py "<trip>\Footage" --apply # -> Horizontal / Vertical / Square
/ Photos
python Tools\spp_sort_media.py --undo "<trip>\Footage\spp_sort_log_<date>.csv"
```

In Resolve (step by step: [SPP_Menu_Scripts.md](#) §A): new project → *Project Settings* → *Master Settings* → *Working Folders* → *Project media location* = `<video>\Resolve Media` → import the footage from `<trip>\Footage` (not the kit folder — the kit is already inside Resolve) → **New Timeline** → **Import Brand Graphics** →

SFX - Import Library → **Music - Import Library**. Import the footage, then run **Moments - Analyse Trip** once per trip (it works in the background — see §15d).

3. New timeline

New Timeline - YouTube 16x9 (3840×2160, scale-to-fit) or **New Timeline - Shorts 9x16** (1080×1920, scale-to-crop). Every timeline gets the same named tracks:

Track	Name	Use
V1	Main (A-Roll)	main story clips
V2	B-Roll / Overlay	cutaways
V3	GFX / Info Cards	SPP titles, map, watermark
A1	Nat Sound / Dialogue	clips' own sound, family talk
A2	VO	your voice-over (the caption sync looks for this name)
A3–A5	Music 1–3	music beds / cross-fades
A6–A8	SFX 1–3	whooshes, ambience
ST1	Subtitles	Hindi subtitles (source for captions + YouTube CC)

4. Tag shot types (colours)

Select clips (timeline or media pool) → **2 Marking & Moments** → **Tag Shot Type** → **1...9**. Sets the clip colour **and** a keyword, so you can make Smart Bins like *Keywords contains "Hero Shot"*. **0 Clear Tag** removes it.

Keyboard shortcuts — run **1 Setup** → **Keyboard Shortcuts - On or Off** once, then tag while you watch:

Key	Tag	Key	Does
4	Hero Shot	9	Reject
5	A-Roll Family Talk	0	Clear Tag
6	B-Roll Scenery	Ctrl+Alt+B	Import Brand Graphics
7	Road Drive	Ctrl+Alt+M	Music - Import Library
8	Kids Moment	Ctrl+Alt+S	SFX - Import Library

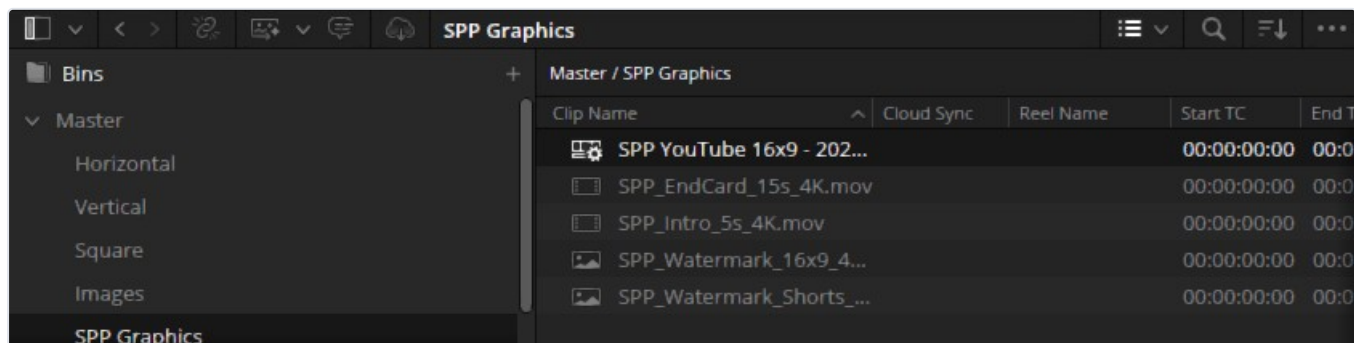
They work only while Resolve is in front on the Media / Cut / Edit page and you're not typing in a text box; 1 2 3 and Q W E are untouched. 4-0 replace Resolve's multicam angle keys (numpad keys untouched).

Ctrl+Alt+Shift+K pauses them; running the script again turns them off. Drone, Timelapse and Stay/Food stay in the menu.

1	Hero shot	Orange
2	A-roll / family talk	Yellow
3	B-roll / scenery	Green
4	Road / drive	Blue
5	Drone / aerial	Teal
6	Timelapse	Purple
7	Kids moment	Pink
8	Stay / food / local life	Tan
9	Reject	Chocolate

5. Brand graphics: intro, end card, watermark

Import Brand Graphics puts them in an **SPP Graphics** bin:



Intro (5 s, transparent)

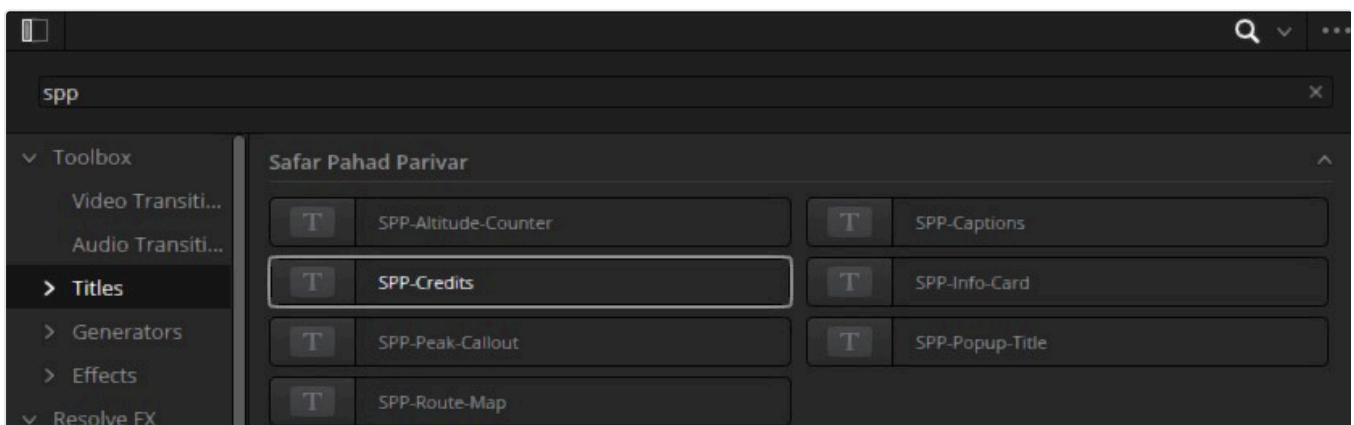
End card (15 s) — drop the next-video / suggested-video end screens on the two boxes in YouTube Studio



The intro ends on "... presents" — what it presents is the film: put an **SPP Film Title** (§10a) straight after the intro, over the first wide shot (or on its own over the dark end of the intro). Intro → Film Title → cold open or Chapter 1.

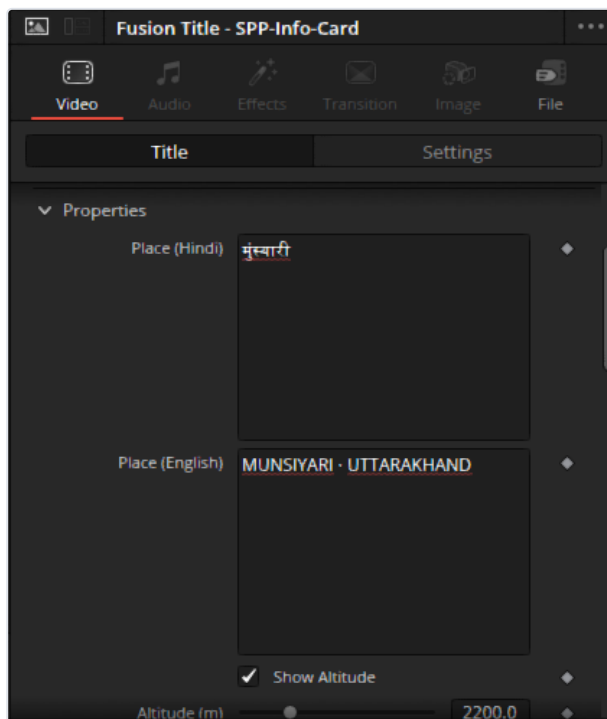
Watermark: `SPP_Watermark_16x9_4K.png` (or `_Shorts_9x16`) on the top video track for the whole film, opacity ~70 %. Brand board (colours, fonts, logo): [Brand/brand_board.png](#) .

6. Titles — how they work



- Drag a title onto **V3**, stretch it as long as you want it on screen. Select it → **Inspector** → **Video** → **Title** shows its settings.
- Text fields accept Hindi directly. Numbers are sliders/boxes; tick-boxes switch parts on/off.
- **Choice fields are typed words** (Resolve 21 shows them as text boxes). The field name lists the choices, e.g. *Position: bottom-left / bottom-right / top-left / top-right* → type `top-right` (just `top-r` or the number `3` also works).
- **Animate Out At (s, 0 = end)** — when the title leaves; 0 = at its end.
- **Changing a title's length** (Ctrl+D, dragging its end): run **4 Titles & Graphics** → **Titles - Follow Clip Length** once after adding titles. Then they behave like Resolve's own titles — entrance fixed at the start, exit always on the last frame, the hold in between stretches (a short Info Card also plays its entrance faster). *Fill from GPS, Chapters* and *Auto Sound for Titles* do it too. Titles can be made shorter, not longer than their default length.
- **Accent Colour** — defaults to brand gold `#F4B03E` .

- Works in 16:9 and 9:16 timelines (layouts adapt automatically).

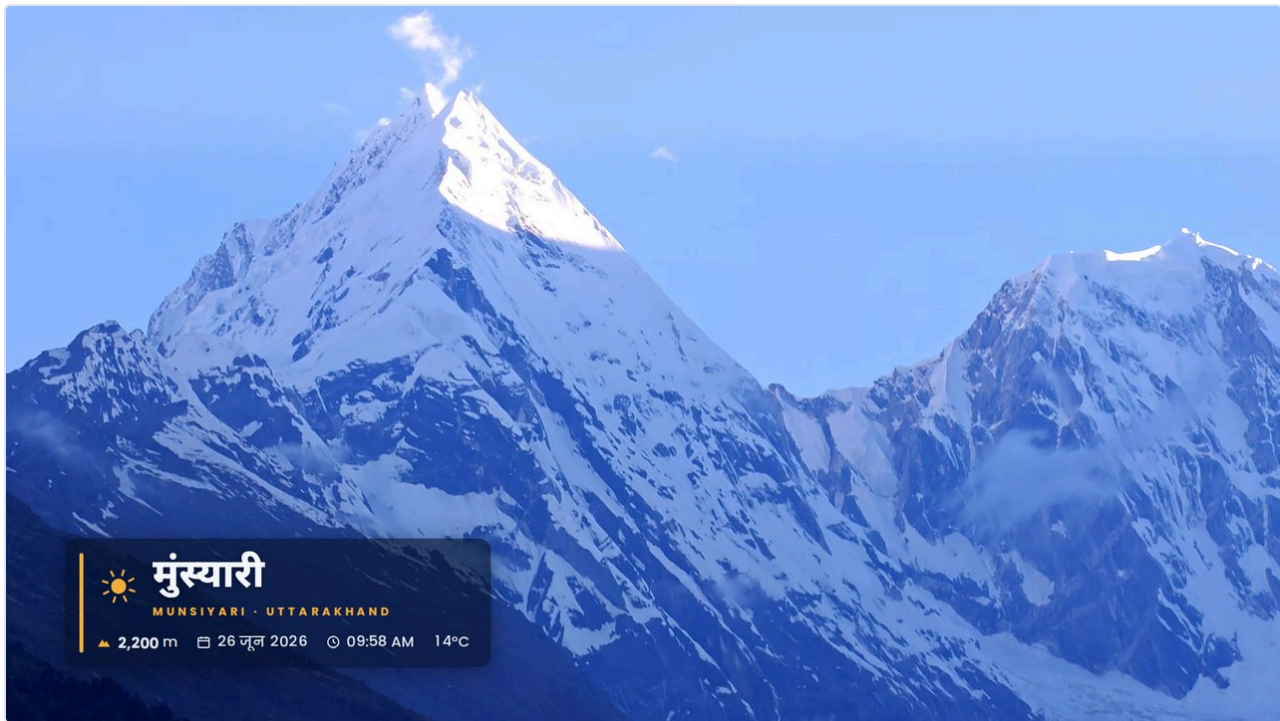


7. Info Card (+ fill from GPS)

It tells its story in order, so the eye follows it: the **gold bar rises** from below → the **panel opens** out of the bar →

where: weather icon, then the Hindi place and the English line slide in from the right → **when / how high:** altitude, date, time and temperature slide up one after another. At the end it leaves in reverse (1 s before the clip ends, or at *Animate Out At*). Hidden items are skipped, so the sequence never has gaps.

Before → now. A card can show the journey since the previous card: the clock runs on from the last card's time (past midnight the date turns over with it), the altitude climbs or drops, the weather icon turns from the old weather to the new — glowing gold while it moves. **Info Cards - Fill from GPS** fills these *From* values from the previous card automatically (the first card has none, so it simply slides in). No reference wanted? Untick *Count on from the previous card* — the values just slide in, fixed. Place · date · time · altitude · weather · temperature. Each item can be hidden.



Setting	Notes
Place (Hindi) / Place (English)	English line looks best in CAPS with <code>· STATE</code>
Show Altitude, Altitude (m), Count Altitude Up	counts up from 0 like an odometer
Show Date / Date, Show Time / Time	type as you want it shown: <code>26 जून 2026</code> , <code>09:58 AM</code>
Weather	<code>none</code> / <code>sun</code> / <code>part-cloud</code> / <code>cloud</code> / <code>rain</code> / <code>snow</code> / <code>fog</code> / <code>night</code>
Temperature	e.g. <code>14°C</code> (blank = hidden)
Position, Size	corner + scale
Rolling-dial numbers	digits spin like a dial (see below); untick for plain numbers

Automatic fill: put Info Cards over your shots, then run **Info Cards - Fill from GPS**. Every card whose *Place (Hindi)* is **empty** is filled from the shot underneath: place (Hindi + English), altitude, date, time, weather, temperature. To refill a card, clear its *Place (Hindi)* and run again. Check Hindi spellings — where OpenStreetMap has no Hindi name you get the English one.

Rolling-dial numbers (Info Card + Altitude Counter):



8. Altitude Counter

Big counter from *Start* to *End* altitude over *Count Duration*, with a small mountain profile.



Tip: use it when the drive climbs (e.g. Dharchula 915 m → Dugtu 3,200 m). Position `centre` + Size 1.4 for a full-screen moment.

9. Peak Callout

Names **up to 4 mountains in one clip** — one title on one track, however many peaks are in the shot. Each summit gets a marker, a white line straight up, a short yellow tick and the name + height right beside it. The peaks appear one after the other.



- **Peak 1 ... Peak 4** — one line each: Hindi | ENGLISH | metres , e.g. पंचाचूली II | PANCHACHULI II | 6904 . Only Hindi is fine too (नंदा देवी). Leave a peak blank to hide it.
- **Peak X / Y** under each — where that summit is, in % of the frame (0,0 = top-left). Scrub to a still frame, estimate, adjust.
- **Line Length** — how high the names sit above the summits (% of the height). Names that would overlap each other or a line are lifted automatically, point away from the middle of the group, and flip side near the frame edge.
- **Marker** dot (pulsing) or arrow (points down at the summit). **Show Heights, Dark Box Behind Names, Size.**
- If the camera moves, keyframe nothing — just keep the callout short (3–4 s) on a steady shot.

10. Pop-up Title

Place / moment title inside a chapter: small kicker, big Hindi headline, English line. For the start of a chapter use the SPP Chapter card (§10b).

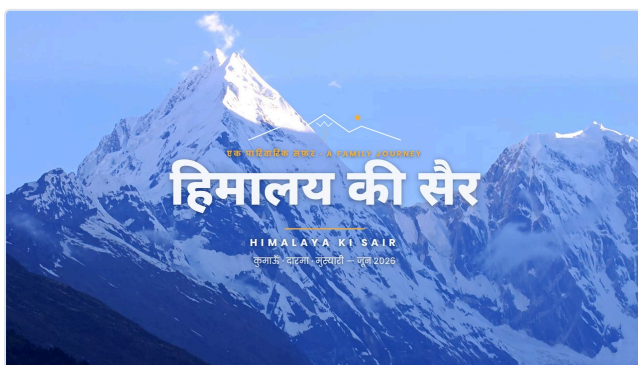


Darken Background adds a soft vignette so the text reads over bright snow.

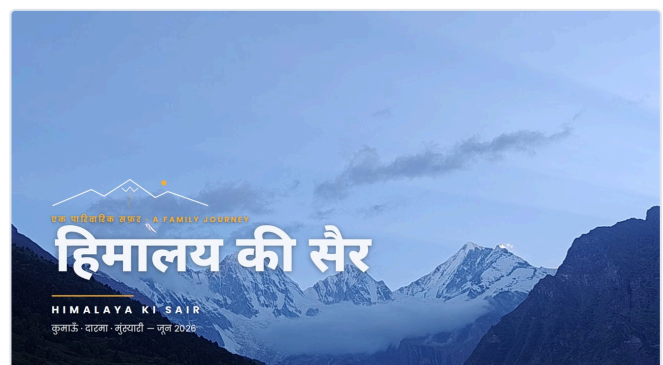
10a. Film Title

The film's name, right after the SPP intro ("... presents"): a mountain line draws itself with the sun and the trail, the Hindi title wipes in, then a gold rule, the English title and a place / date line.

Centre (over a wide shot)



Bottom-left (over action)



Fields: *Small line above* (e.g. एक पारिवारिक सफ़र • A FAMILY JOURNEY), *Film Title (Hindi)*, *(English)*, *Place / date line*, *Draw the mountain line*, *Position: centre / bottom-left*, *Darken Background*, *Size*, *Animate Out*. Length: 7 s (stretch or shorten the clip; the picture slowly pushes in while it's on screen).

10b. Chapters

Long films read better in chapters — and on YouTube the chapter list splits the progress bar so viewers can jump.

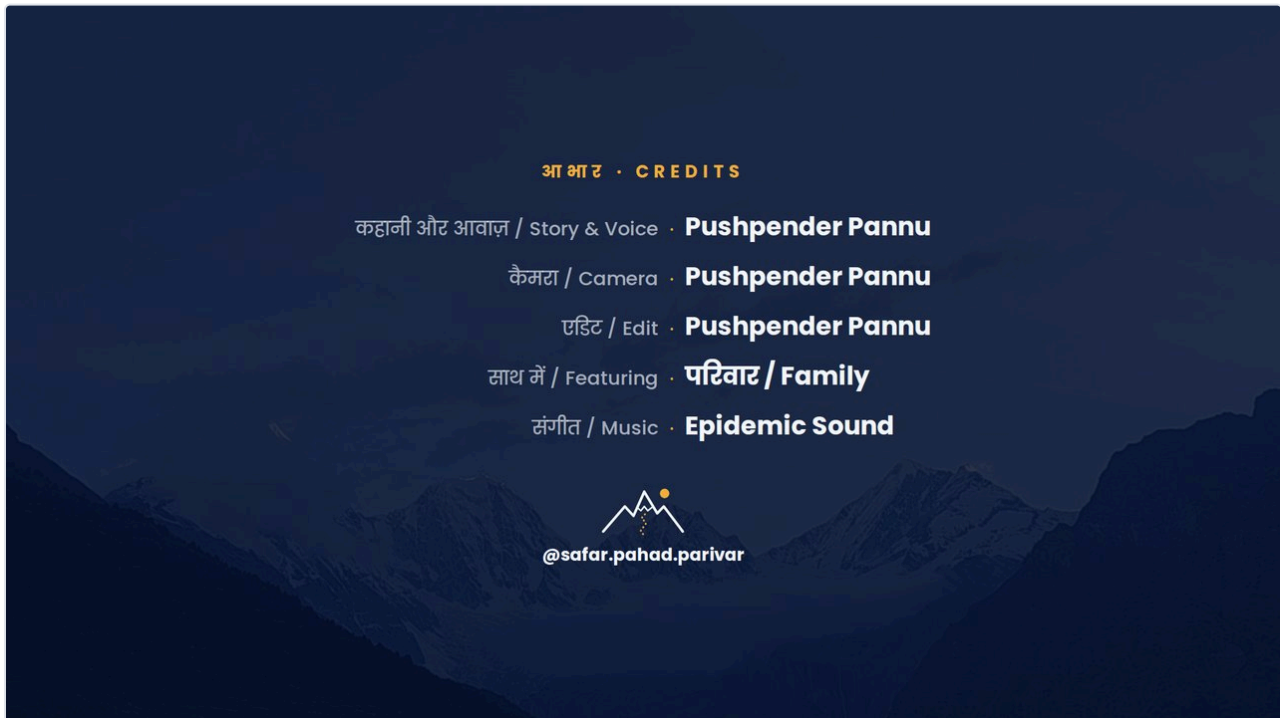


1. At the start of each chapter put an **SPP Chapter** title on V3 (~5 s) over the chapter's first shot.
2. Type *Chapter name (Hindi)* and *(English)*, and a *Day / route line* (दिन 2 · धारचूला → दुग्त्) — or leave it blank. *Position*: centre (big moment) / bottom-left / top-left (over action). *Hindi digits* shows ०२ instead of 02.
3. Run **4 Titles & Graphics** → **Chapters - Number + YouTube List**: the chapters are numbered in order, each card's trail shows *chapter X of N*, a purple marker is put on each, and the YouTube list (00:00 शुरुआत · Intro , 01:12 अध्याय 1 · ...) is copied to the clipboard and saved in <trip>_spp_youtube\ . Paste it into the description. Run it again after every edit (and once more after the final render). YouTube only shows chapters when there are **3 or more**, the first at **00:00**, each **10 s or longer** — the script adds the 00:00 intro line and warns about the rest.



11. Credits

Heading + up to 7 lines written Role | Name (blank lines are hidden), logo and handle.



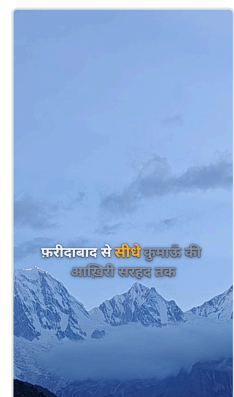
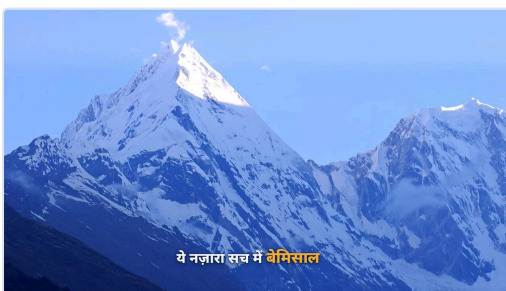
12. Hindi captions (+ sync to VO)

Resolve's own *animated* subtitles break Hindi (matras/conjuncts come apart). **SPP Captions** animates whole words, so Hindi always shapes correctly.

Word pop + gold highlight (pop)

Karaoke sweep (karaoke)

Shorts (raised)



Recommended — synced to your voice (one click):

1. Subtitles on the subtitle track (type them, or *Timeline* → *Create Subtitles from Audio* and correct the text). Put ***stars*** around a word to force a stress: ये नज़ारा ***बेमिसाल*** था (stars never show).
2. VO clips on the audio track named **VO**.
3. One **SPP Captions** title on V3 over the VO section.

4. Run **Captions - Sync Words to VO**. Resolve pauses (~1 min per 10 min of VO). It finds when every word is spoken, trims silences, marks stressed words (louder/stretched → bigger, gold) and fills the caption clip (text + start time).
5. **Turn the subtitle track off** (its eye/enable button) so captions aren't shown twice. Keep it — export it as SRT for YouTube CC (*File → Export → Subtitle*).

Changed the subtitles or re-cut the VO? Run the sync again.

Manual (no sync): *File → Export → Subtitle → SRT*, open it in Notepad, copy all, paste into **SRT text**, and set

This clip starts at timeline time (s) (clip at 00:02:15 → 135). Words then spread evenly across each line.

Settings: **Animation** pop / karaoke / fade , **Position** bottom / raised / centre / top , Size, Dark plate, Text + Highlight colour.

13. Route Map

Relief map in brand colours, dotted path along the **real roads**, stops pop up with arrival/departure times, a running date-time clock, camera following the journey.





1. Put an **SPP Route Map** title on V3, 12–20 s.
2. Run **Route Map - Build from Timeline**. It uses only the dates of the footage on this timeline, builds the map (1–3 min first time) into `<trip>\Route Maps\<timeline name>\` and loads it into the title.
3. **Fix / add stops**: open `stops.csv` in that folder (Excel). Columns: *hindi, english, place, arrive, leave*. Fix names, delete rows, or add a stop GPS missed — the place can be a name (Dharchula) or `lat, lon` ; times as `2026-06-23 18:00` . Save → run the script again.
4. Inspector: *Title, Route starts drawing at / finished at, Pause at each stop, Camera whole / follow, Follow zoom*, times on/off, running clock on/off, keep earlier names, darken map. - A Shorts (9:16) timeline gets a portrait map automatically. - Keep the credit line (bottom-right) — the map data licences require it. - `route.gpx` in the same folder can be loaded into 3D fly-over apps (see `Docs\Maps_GPS_and_Terrain.md`).

Better paths: export your Google Maps Timeline (phone: Maps → profile → Your Timeline → ⋮ → Location & privacy settings → Export Timeline data) and save `Timeline.json` in the trip folder, then:

```
.\Tools\.venv\Scripts\python.exe Tools\spp_gps.py index "<trip>" --timeline "<trip>\Timeline.json"
```

14. GPS lookup tool

The menu scripts above use it; you can also ask it directly (PowerShell in the kit folder):

```
$py = ".\Tools\.venv\Scripts\python.exe"
& $py Tools\spp_gps.py at "<trip>" "2026-06-26 11:14" # where were we? place,
altitude, weather
& $py Tools\spp_gps.py file "<trip>" VID20260626111444.mp4 --offset 12
& $py Tools\spp_gps.py stops "<trip>" --csv stops.csv
& $py Tools\spp_gps.py gpx "<trip>" trip.gpx
& $py Tools\spp_gps.py serve "<trip>" # http://127.0.0.1:8777/at?t=2026-06-26T11:14
(JSON)
```

Example answer: Sela · 2,378 m · 22°C · drizzle . Full details: [Maps_GPS_and_Terrain.md](#) .

15. VO sound recipe

On the **VO** track (Fairlight): Voice Isolation 60–70 → EQ (high-pass 80 Hz, small cut 250–350 Hz, lift 3–5 kHz) → De-esser → Dialogue Leveler / compressor 3:1 → final mix **–14 LUFS**, music 15–20 dB under the voice.

15a. Sound effects

A brand sound library made for the SPP titles — **269 sounds, 846 files** (incl. real field recordings) in three styles plus cinematic risers, impacts and music transitions, several variations each, seamless loops for any length.

Strings (default): real recorded orchestra — violin, viola and cello sections, solo violin, harp, timpani, gong, Nepalese bells; warm and emotional, all in D major so the sounds fit together. **Grand**: deep cinematic synth — sub-bass hits, taiko/dhol, braams, gongs, ransingha-style horn with valley echo, drone beds. **Light**: playful UI ticks, pops and marimba. Full list: [SFX_Library.md](#). Listen first:

SPP_SFX_RealWorld_Demo.mp3 , SPP_SFX_Strings_Demo.mp3 , SPP_SFX_Cinematic_Demo.mp3 ,
SPP_SFX_Grand_Demo.mp3 , SPP_SFX_Demo_Reel.mp3 .

Build it once per PC: `.\Tools\make_sfx.ps1` → `<kit>\SFX\`. The first run downloads the recorded strings (*VSCO-2 Community Edition*, public domain / CC0 — free for YouTube, no credit needed; ~1.1 GB into `Source\_vsc` incl. the music instruments, not in git) and takes ~10 min; later runs ~3 min. Then **SFX - Import Library** puts it in an **SPP SFX** bin. The sounds are built from code + those recordings, so they are identical on every PC and don't need backing up.

Folder	What's in it
01 Title Kits	sounds timed to each SPP title (Info Card, Altitude 1.5–6 s, Peak, Pop-up, Credits, Route Map, title-out, caption tick)
02 UI	pops, soft ticks, mouse click, notification bell, arrival chimes, shimmer, swipes
03 Motion	whooshes (short/medium/long, in/out), swish pans, risers 2/4/6 s, reverse swell, soft impact, cinematic boom
04 Dial & Mechanics	dial ticks, odometer rolls 1–6 s (slow down and settle like the rolling numbers), digit spin-stop, gear loop , clock tick-tock loop, time-lapse clock loop, ratchet
05 Map & Travel	map unfold/fold, pencil-drawing loop, dotted-trail loop, pin drop, map zoom in/out, travel-motion loop, camera shutter
06 Bells & Brand	mandir bell, hand ghanti, wind chime, intro sting and end-card sting timed to the SPP intro / end card
07 Nature Beds	30 s seamless: mountain wind (calm/gusty/high whistle), prayer flags, mountain stream, light rain, night crickets
08 Phone	Indian ringback / dial / busy tones, ringtones, vibrate, keypad dialing, pickup, hang-up, call-ended beeps, message ping, line noise
09 Grand Title Kits	grand versions of every title kit (Info Card, Altitude 1.5–6 s with 3 landings, Peak, 5 chapter-title hits, Credits, Route open, stop hits, journey pulse loop, title-out)
10 Grand Impacts & Swells	braams, taiko hits, sub booms, valley-echo boom, deep whooshes, riser-into-hit, deep reverse swells
11 Grand Bells & Horns	gong / tam-tam, deep mandir bell, singing bowl, ransingha horn calls with valley echo
12 Grand Drones & Drums	30 s drone beds (warm, hopeful, dark, airy), dhol-damau pulse loops, heavy gears, vast mountain air

Folder	What's in it
13 Strings Title Kits	strings versions of every title kit: harp + pizzicato Info Card, tremolo that climbs with the altitude count and lands on timpani + chord, harp-to-violin peak, 5 chapter-title hits, credits swell, route open, pizz + bell stop hits, spiccato journey loop, title-out
14 Strings Hits & Swells	section swells, tremolo risers, string stabs, timpani hits and rolls, gong, bass drum, cymbal swells, solo-violin phrases (pahadi-style pentatonic)
15 Strings Beds & Bells	30 s string pads and cello drone loops, harp glissandos, Nepalese bells, pizzicato pops
16 Cinematic Risers	Riser_Epic 3/5/8/12 s (tremolo strings, timpani + snare roll, cymbal → BIG HIT exactly at 3/5/8/12 s) · Riser_Awe 3/5/8/12 s (for waterfalls and big mountains: strings bloom, harp sweep, opens into a warm chord) · Riser_Tension 3/5/8 s (suspense, cuts to silence)
17 Downers & Sub Drops	sub drops, boom + sub drop (with valley echo), downers (strings slide down, energy drains), power-down / tape-stop
18 Impact Drums	big hits (taiko + timpani + bass drum + strings/crash/gong/boom), drum fills that run into a hit, soft hits, valley-echo hits
19 Music Transitions	for all 24 keys : Swell_Into_<key> ×3 (strings+cymbal / +harp / soft — lands on the new music), Tail_<key> ×2 (covers an early music cut); Bridge_<a>_to_ between the SPP music keys (D, E, Bm, Dm) — any other pair is made on demand by Music - Key Transition
20 Rain & Weather	REAL recordings: light / heavy rain, rain on roof / tin roof / the car (inside), forest rain, thunder, strong wind
21 Car & Road	REAL: car doors open / close / handle / lock / boot, engine start, horn, pass-bys (dry and wet), wet-road traffic, tyres on gravel, footsteps on gravel
22 Birds	REAL: cuckoo calls + morning beds, songbird whistles, crows, raven, pheasant, kite / buzzard, dawn chorus, Indian forest birds
23 Water	REAL: close streams (one from Nepal), rivers, riverside with birds, waterfall close / distant, night river with crickets, drips, splashes
24 Temple & Town	REAL: conch (shankh), temple bells, mountain horn, monastery chant, Indian crowd / town, wedding and street dhol

One click for all titles — SFX - Auto Sound for Titles. Pick a style — **Strings** (default), **Grand**, **Light** or **Mix** (grand for chapter titles, altitude, route map, intro/end; light for small cards). It finds every SPP title (and the SPP intro / end-card clips) on the timeline and lays the matching sounds on the **SFX 1–3** tracks, exactly on the animation beats: card whoosh, row pops, odometer roll that slows with the numbers, pin drop + chime at each route stop (times read from the route file), dotted-trail loop while the path draws, out-whoosh. Variations rotate so repeats never sound the same. Auto clips are **Lime**; run again after editing — it replaces the old Lime clips. Keep a tweaked sound by changing its clip colour. **SFX - Remove Auto Sounds** clears them.

Land it on the cut — SFX - Land at Playhead. Put the playhead on the reveal / title / first beat of the next track, select a sound in the Media Pool (a riser, **Swell_Into_***, **Bridge_***, drum fill...) and run it: the sound is placed so its big moment lands exactly on the playhead (a **Riser_Epic_8s** starts 8 s earlier). Every sound's landing time is stored in the library.

Changing music — Music - Key Transition. Select the two music clips on the timeline (the one ending and the one starting) and run it. It listens to the end of the first and the start of the second, shows their keys (e.g. *D* → *E*, with the runner-up — relative keys like *D* / *Bm* share their notes, either works), lets you change them, then places a **Bridge** (old key → new key), a **Swell into** the new key, or **Tail + Swell** on a free

SFX track so its big moment lands exactly on the first frame of the new music. One clip selected: swell into it, or tail at its end. Works on any music (Epidemic Sound too). Command line: `Tools\spp_key.py <file> -start 60 --dur 20 (find a key) · Tools\spp_key.py render bridge Ab Em --out x.wav (make a transition)`.

Near or far — Distance - Selected Clips. You don't need separate near/far files. Select audio clip(s) on the timeline, run it, pick *Near (~10 m) · Mid (~50 m) · Far (~200 m) · Very far (~600 m) · Across the valley (echoes)*. It makes a processed copy on a free track and switches the original off (D to switch back). What it does — and what you'd do by hand in Fairlight: cut the treble (air absorbs it: high-cut ~8 kHz at 50 m, ~4 kHz at 200 m), cut the low bass (no proximity), lower the level, add more reverb than direct sound (reverb send, longer decay), narrow the stereo (a far sound is a point), and for valleys add 2–4 darker echoes 0.4–2 s apart. Tick *Keep the loudness* if you'd rather set the level with the fader. Loops stay seamless. Command line: `Tools\spp_distance.py <file> --preset far`.

Any length — SFX - Loop Fill (In to Out). Select one `_LOOP_` sound in the media pool, set In/Out on the timeline (I / O), run it: the loop is laid end-to-end with seamless joins (last copy trimmed). Great for wind, stream, gear, clock, dotted trail.

Mixing tips: title sounds sit around –6 dB under the VO already; ride the SFX tracks down (–6 to –12 dB) when the voice is on. Beds (wind, stream) at –18 to –24 dB under dialogue.

15b. Phone-call voice

Select one or more **audio** clips (a VO line, or someone's dialogue) → **Phone Voice - Selected Clips** → choose a style (`mobile`, `landline`, `speaker`, `walkie`, optional line hiss / tiny network glitches). A processed copy is placed at exactly the same spot on a free audio track and the original clip is switched off (select it and press **D** to switch it back on). Add the phone sounds around it: `SPP_Ringtone` / `SPP_Vibrate` → `SPP_Pickup` → (*phone voice*) → `SPP_Hangup` + `SPP_Call_Ended_Beeps`; for the caller's side, `SPP_Keypad_Dialing` → `SPP_Ringback_India`.

Command line (any audio/video file): `& $py Tools\spp_phone_voice.py "<file>" --style mobile --noise`

Live alternative inside Resolve (Fairlight, adjustable): on a track holding only the phone lines add

EQ: high-pass 300 Hz (steep), low-pass 3.4 kHz (steep), bell +4 dB at 1.8 kHz → **Dynamics:** compressor 4:1, threshold ≈ –25 dB → optional **Distortion** (small amount) for a cheap handset. Speakerphone: add a short small-room **Reverb**.

15c. Background music

Twelve original background tracks (2–3 min each) composed in code and played with recorded instruments — real strings, flute, upright piano, harp, glockenspiel, timpani, gong and Nepalese bells (VSCO-2 Community Edition, CC0) — plus a synthesised tanpura, temple bells, manjira and mountain wind. Listen first: `Docs\SPP_Music_Preview.mp3` (32 s of each). Full list: [Music_Library.md](#).

Theme	Tracks
01 Slow Build	First Light (piano → full strings, sunrise/openings) · Long Road Up (driving ostinato, drives/climbs) · Summit Rise (slow epic rise, peaks/reveals)
02 Temple Bells	Mandir Dawn (tanpura, bells, low bansuri) · Aarti Glow (manjira, hand bells, devotional tune) · Himalayan Gongs (no rhythm: gongs, bowls, drone)

Theme	Tracks
03 Wind	High Pass (wind + airy strings) · Prayer Flags (fluttering harp/glock + bright flute) · Snowline (cold wind, lonely viola)
04 Flute	Bansuri Valley (alaap then Bhupali tune) · Pahadi Dhun (6/8 folk tune) · Evening Raag (Raag Yaman, sunsets)

Build it once per PC: `.\Tools\make_music.ps1` (≈10–15 min; uses the same instrument download as the sound effects) → `<kit>\Music\`. Then **Music - Import Library** puts the tracks in an **SPP Music** bin (Purple clips). Rebuild one track: `.\Tools\make_music.ps1 Pahadi`.

Using it: put music on the *Music* track, duck it under the VO (–18 to –24 dB while the voice talks, –10 to –12 dB alone). Tracks start soft and end on a proper final chord / bell, so you can cut from the middle and let the ending ring out.

Rights: everything is original (our code) or public domain (CC0 recordings), so it is free to use on monetised YouTube with no credit needed. Don't register these tracks with a Content-ID service — that could cause claims on your own videos. (The upright-piano recordings are by Simon Dalzell / Ivy Audio via Versilian Studios; credit is welcome, not required.)

15d. Moment finder

Finds the family moments in a whole trip's footage — laughter, kids shouting or cheering, singing, "wow / papa dekho" reactions, names being called (Pihu, Oju, Meenakshi, Brijesh, Alka, Pushpender) — and writes a Hindi transcript of everything that was said.

- Moments - Analyse Trip** (runs in the background; a 2-hour trip takes ~20–40 min the first time, later only new clips).
- Read `<trip>\_spp_moments\Moments.md` : **3 opening candidates** for the cold open, the top 40 moments, and a day-by-day list. `Transcript.md` has every sentence with clip name and time (Ctrl+F for पानी, बर्फ, पिहू...).
- Moments - Add Markers:** coloured markers on the source clips — Yellow laughter, Pink kids talking, Red shouts, Fuchsia cheering, Purple singing, Green reaction words (the marker note holds what was said).
- Moments - Best Moments Timeline:** a new timeline of the top 20 / 40 / 80 moments, only laughter, only kids, or the 3 openers, in the order they happened.
- Moments - Search Transcript:** type words (comma-separated) → a timeline of every clip where they were said. Engines: faster-whisper large-v3 (speech, on the GPU) and PANNs (AudioSet sound events). Automatic Hindi transcripts have mistakes — use them to *find*, not to quote.

15e. Reflections and glare through the car window

Two different problems, two tools:

On screen	What it is	Tool
Shapes on the glass: the dashboard and its vents, the phone, a hand, a white shirt	reflection — stays in the same place while the road slides past	menu script Reflection - Clean Selected Clips

On screen	What it is	Tool
Everything low in the frame looks milky / washed-out	glare — sunlight on the dashboard or dusty glass adds a veil	colour tool SPP Glass Glare (DCTL)

Resolve's own **Dehaze** (ResolveFX Refine) works on the glare, not on the shapes; SPP Glass Glare does the same job but is aimed at the lower frame where windshield glare sits, keeps whites white and has a map to see what it does.

Workflow

1. New trip: **Reflection - Scan Trip** (background) → run again → Orange/Yellow clips in the Media Pool = check these.
2. Put the shots on the timeline, trim them, select → **Reflection - Clean Selected Clips** → *Preview pictures only* to check → run again → **Clean** (background, ~4–5 min per minute of 4K) → when the window says *All done*, run it once more.
3. The cleaned version is a **second take** of the same clip (teal). **Reflection - Show Original or Cleaned** flips every cleaned clip for a before / after.
4. Grade: **SPP Glass Glare** on the first node (Glare amount 0.08–0.12 to start; *Show glare map* to aim it).

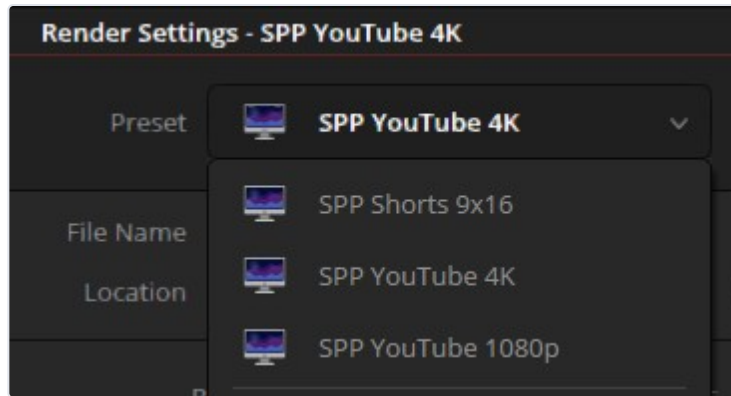
How the cleaner works: it looks at up to a minute of the clip. Scenery changes, the reflection doesn't — whatever is still there in every part of that minute, over areas where the landscape keeps rushing past, is the reflection. It takes that off in linear light (like peeling off the light the glass added), follows how strong it is second by second (the sun on the dashboard changes), and leaves alone anything that isn't moving behind the glass: the bonnet, the window frame, the sky, the far view straight ahead.

Better at the shoot (saves more than any fix) — also on the *Clothing Colours* print-out:

- **Black non-slip dash mat** — the dashboard can't reflect what's black and matte.
- **Silicone anti-reflection lens hood** (rubber cone that presses against the glass) — blocks the car's inside.
- **Phone lens flat against the glass**, not angled; clean the glass inside and out; wipe the lens.
- **Clip-on polarising filter (CPL)** — turn it until the glare on the glass disappears (also deepens the sky).
- Front-seat passengers in **dark tops** (bright clothes reflect in the windshield — the opposite of the outdoor rule).
- Film from the **shady side** of the car; tilt the phone slightly down; switch off the phone screen's reflection by dimming its brightness.
- Side windows reflect less than the windshield; an open window reflects nothing (when it's safe).

16. Render (Deliver)

Pick a preset: **SPP YouTube 4K**, **SPP YouTube 1080p**, **SPP Shorts 9x16**.



Save the final file into `<video>\Exports` . Then *File* → *Export Project* (.drp) into `<video>\Resolve` .

17. Backup with git

- The kit is in git (GitHub, private). After changes: `git add -A` , `git commit -m "..."` , `git push` .
- Footage is **not** in git — back up `F:\Video Editing\Projects` to an external drive / cloud.
- Keep music/SFX used in a video inside its `Audio` folder.

18. Troubleshooting

Problem	Fix
New titles/scripts don't appear	Run <code>.\install.ps1</code> , restart Resolve.
Title shows old settings after a kit update	Delete the clip and drag a fresh one from Effects.
Captions show twice	Turn off the subtitle track.
Hindi letters broken in captions	You used Resolve's animated subtitles — use SPP Captions.
"Word-timing engine not installed"	<code>.\Tools\setup_word_timing.ps1</code>
Route map shows "Choose route.json"	Run Route Map - Build from Timeline , or pick <code>route.json</code> in the Inspector.
Route has few stops / straight lines	Add stops in <code>stops.csv</code> ; add a Google Timeline export.
Info Card not filled	Its <i>Place (Hindi)</i> wasn't empty, or the shot has no GPS near that time (add Timeline export).
"The SFX library isn't built yet"	<code>.\Tools\make_sfx.ps1</code> , then SFX - Import Library .
Auto sounds in the wrong place	Titles moved after running it — just run SFX - Auto Sound for Titles again.
Resolve froze while a script ran	Normal for caption sync / route map / key detection — wait; the Console (Workspace → Console) shows progress.
"Couldn't find the trip folder" (Moments)	Import some of the trip's footage (from <code><trip>\Footage</code>) into the project first.

Problem	Fix
Moments: only speech, no laughter / cheering	The sound-event model isn't downloaded — run <code>.\Tools\setup_word_timing.ps1</code> again, then Moments - Analyse Trip (it only adds what's missing).
Moment analysis is very slow	It fell back to the CPU — check <code><trip>_spp_moments\analyse_log.txt</code> for "GPU not available"; re-run <code>setup_word_timing.ps1</code> .
Key Transition picked the wrong key	Change it in the window before placing. Relative keys (D / Bm) share notes — either sounds right.
No folders 20–24 in the SFX library	<code>Tools\freesound_key.txt</code> missing when <code>make_sfx.ps1</code> ran — add the key and run it again.
YouTube doesn't show chapters	3+ chapters, first at 00:00, each ≥ 10 s — run Chapters - Number + YouTube List and read its notes.
No <i>SPP Film Title</i> / <i>SPP Chapter</i> in Effects → Titles	Run <code>.\install.ps1</code> , restart Resolve (titles are only read at start-up).
A title vanishes at its end instead of animating out	Run Titles - Follow Clip Length (4 Titles & Graphics) once — after that its exit follows the clip end.
Titles look twice too big on a 4K timeline / a peak name far from its line	Fixed in the Sept-30 templates (run <code>.\install.ps1</code> , restart Resolve). If you had set Size 0.5 to compensate, set it back to 1 .
A menu script "disappeared"	They're in folders now: Workspace → Scripts → Safar Pahad Parivar → 1 Setup ... 6 Sound & Music.
Reflection cleaning darkened the landscape / blotches	Run Reflection - Clean Selected Clips again with <i>Gentle</i> , or switch that clip back with <i>Show Original or Cleaned</i> .
Cleaned take is shorter than the clip after trimming	Run Reflection - Clean Selected Clips again — it cleans the new range.
No <i>SPP Glass Glare</i> in the DCTL list	Run <code>.\install.ps1</code> (as administrator if it says it couldn't copy), restart Resolve. DCTL needs Resolve Studio.
A sound is missing in the bin after a rebuild	Run SFX - Import Library / Music - Import Library again (only new files are added).